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ADJUSTABLE EQUIPMENT STORAGE RACK FOR A BOAT

Abstract

An equipment storage rack for a boat can store a piece of equipment on the exterior of the boat and can be adjustable. For example, the equipment storage rack can be attached to an upper perimeter surface of a barrier of the boat and can be adjustable for different widths of the upper perimeter surface. The adjustable equipment storage rack includes a cradle for supporting a piece of equipment, and a mounting bracket connected to the cradle to support the cradle. The mounting bracket has a plurality of engagement positions for a fastener to engage with the mounting bracket and to secure the mounting bracket to the boat at a fastening position. The cradle is a different distance from the fastening position at each engagement position of the plurality of engagement positions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 63/562,897, filed Mar. 8, 2024, and titled “WATER-SPORT EQUIPMENT STORAGE RACK FOR A BOAT,” the entirety of which is incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present disclosure generally relates to an apparatus for storing equipment on a boat, particularly, watersport equipment on recreational boats used for watersports.

BACKGROUND OF THE INVENTION

[0003] Watersport boats often are used to tow watersport participants, such as water skiers, wakeboarders, and the like, using a tow-line. For water skiing and wakeboarding, the participant holds onto one end of the tow-line and the other end is attached to the boat. For tubing, the tow-line is attached to the tube and the watersport participant(s) holds onto the tube. A boat may also be used to generate a wake on which a watersport participant, such as a wake surfer or foiler, may wake surf or foil, generally without holding onto a tow-line once they get going. Another water activity that may be enjoyed with recreational boats is paddleboarding, such as stand-up paddleboarding, which involves standing up on a relatively large board (i.e., a paddleboard) and using a paddle to make way through the water.

[0004] Buoyant water mats, which are also known as water pads, lily pads, or floating islands, are mats that float on the surface of a body of water, such as a pond, lake, or river, and can support one or more persons. Water mats are typically large, rectangular mats that may have a width of 6 feet and a length longer than 6 feet, such as, for example, 12 feet, 15 feet, or 18 feet. With such a large, buoyant area, these mats have a capacity to hold a substantial amount of weight and can suitably hold multiple people at once, such as, for example, four, six, or eight adults. These water mats may be rolled up from their flat, rectangular deployed position in order to be stowed.

SUMMARY OF THE INVENTION

[0005] In one aspect, the invention relates to an equipment storage rack for a boat. The equipment storage rack can be attached to a barrier and can be adjustable relative to the barrier.

[0006] In another aspect, the invention relates to an equipment storage rack for a boat that can store a piece of equipment on the exterior of the boat. The equipment storage rack can be attached to an upper perimeter surface of a barrier of the boat and adjustable for different widths of the upper perimeter surface.

[0007] In a further aspect, the invention relates to an adjustable equipment storage rack for a boat. The adjustable equipment storage rack includes a cradle for supporting a piece of equipment, and a mounting bracket connected to the cradle to support the cradle. The mounting bracket has a plurality of engagement positions for a fastener to engage with the mounting bracket and to secure the mounting bracket to the boat at a fastening position. The cradle is a different distance from the fastening position at each engagement position of the plurality of engagement positions.

[0008] In still another aspect, the invention relates to a boat including the equipment storage rack of any of the preceding aspects.

[0009] In yet another aspect, the invention relates to a boat including a barrier and an adjustable equipment storage rack mounted to the barrier. The adjustable equipment storage rack includes a cradle assembly. The cradle assembly includes a cradle for supporting a piece of equipment and a mounting bracket connecting the cradle to the barrier. The mounting bracket is positionable in a plurality of cradle positions and, at each position of the plurality of cradle positions, the cradle is a different distance from the barrier.

[0010] These and other aspects of the invention will become apparent from the following disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 shows a boat equipped with an adjustable equipment storage rack.

[0012] FIG. 2 shows a portion of the starboard side of the hull of the boat of FIG. 1 with the adjustable equipment storage rack connected thereto.

[0013] FIGS. 3A and 3B are stern views of a barrier with the adjustable equipment storage rack attached thereto. FIG. 3A shows a fastener engaged with a mounting bracket at a second engagement position and FIG. 3B shows the fastener engaged with the mounting bracket at a fourth engagement position.

[0014] FIGS. 4A and 4B are perspective views of a cradle assembly. FIG. 4A shows a front side of the cradle assembly and FIG. 4B shows a back side of the cradle assembly.

[0015] FIG. 5 is a detail view of the mounting bracket of the cradle assembly shown in FIGS. 4A and 4B.

[0016] FIG. 6 is a detail view of a barrier of the boat with the cradle assemblies of the adjustable equipment storage rack removed.

[0017] FIGS. 7A and 7B show a fastener assembly that can be used with the adjustable equipment storage rack. FIG. 7A shows a quick-release fastener connected to a receiver and FIG. 7B shows the quick-release fastener removed from the receiver.

[0018] FIGS. 8A to 8D illustrate the process of adjusting the position of the cradle assembly from the cradle position shown in FIG. 3A to the cradle position shown in FIG. 3B. FIG. 8A shows the mounting bracket and fastener engaged to position the cradle assembly in the cradle position shown in FIG. 3A. FIG. 8B shows a step of disengaging the fastener. FIG. 8C shows a step of moving the mounting bracket. FIG. 8D shows a step of engaging the fastener with the mounting bracket to position the cradle assembly in the cradle position shown in FIG. 3B.

[0019] FIGS. 9A and 9B are perspective views of another cradle assembly that can be used with the adjustable equipment storage rack. FIG. 9A shows a front side of the cradle assembly and FIG. 9B shows a back side of the cradle assembly.

[0020] FIGS. 10A and 10B are perspective views of another cradle assembly that can be used with the adjustable equipment storage rack. FIG. 10A shows a front side of the cradle assembly and FIG. 10B shows a back side of the cradle assembly.

[0021] FIG. 11 is a detail view of a clasp of a strap for the cradle assembly shown in FIGS. 10A and 10B.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0022] As noted above, recreational boats may be used for watersports and other water-based recreational activities. These water activities use various equipment, such as boards (for example, paddleboards, wakeboards, and surfboards), water skis, or water mats, which are referred to generically herein as water-sport equipment. When not in use, this watersport equipment is often stored on the boat, preferably in racks or compartments. Preferably, these racks and compartments are designed to maximize the space onboard the boat and prevent the equipment from moving around while stored.

[0023] The present disclosure generally relates to recreational boats and an apparatus that can be connected to or attached to the boat for storing watersport equipment. More specifically, the apparatus for storing watersport equipment is a rack that can be positioned on the boat to store the watersport equipment on an exterior of the boat, maximizing the space within the interior of the boat. In addition, the rack can be an adjustable rack capable of being attached to surfaces and

features, such as barriers on the boat, of different widths.

[0024] FIG. 1 shows a boat **100** having an adjustable equipment storage rack **200**. FIG. 1 is a perspective view of the boat **100**, showing the stern of the boat **100**. The boat **100** of this embodiment includes a hull **110** having a bow **111**, a transom **113**, a hull bottom **115**, a port-side hull **117**, and a starboard-side hull **119**. The hull bottom **115** includes the portions of the hull **110** between the chines. The boat **100** has a longitudinal centerline **102** running down the middle of the boat **100**, halfway between the port-side hull **117** and the starboard-side hull **119**. The boat **100** shown in FIG. 1 is a bowrider having both a bow seating area **122** positioned in the bow **111** of the boat **100** and a primary seating area **124** (sometimes also referred to as the cockpit) positioned aft of a windshield (not shown). Within the boat's interior **120** is a control console **152** for operating the boat **100**. Here, the control console **152** is positioned on the starboard side of the boat **100**, proximate to and aft of the windshield. Although described in reference to a bowrider, the embodiments discussed herein may be used with any suitable boat including cuddies, center consoles, and cruisers, for example. Various embodiments discussed herein may also be suitable for use with other boats such as pontoon boats.

[0025] Collectively, the bow **111**, the transom **113**, the port-side hull **117**, and starboard-side hull **119** define an interior **120** of the boat **100** with an exterior **130** of the hull **110** being the space outward from the hull **110**. The port-side hull **117** and the starboard-side hull **119** have a port gunwale and a starboard gunwale, respectively, and in some embodiments, the boat can include a port coaming and a starboard coaming formed on the port gunwale and the starboard gunwale, respectively. The hull side (e.g., the port-side hull **117** or the starboard-side hull **119**) and, when present, the coaming, form a perimeter **132** of the boat **100** with a top surface **134** that is generally horizontal. As depicted, this top surface is a generally horizontal deck surface, but this surface can also be formed as part of the hull. This surface is referred to herein as a top perimeter surface of the deck or hull.

[0026] The boat **100** can include a swim platform **106** connected to the stern to make it easier for people to get into the water from the boat **100** or into the boat **100** from the water. The swim platform **106** can be horizontal and should be capable of supporting a human and preferably is capable of supporting at least 500 lbs., and, even more preferably, 1250 lbs. The swim platform **106** can be constructed from any suitable material that can be used in a marine environment including, for example, fiberglass or teak. The swim platform **106** can be integrally formed with the stern of the boat **100** (as shown in FIG. 1), but the swim platform **106** also can be an attachable/detachable platform that is attached to the transom **113** of the boat **100** using, for example, two brackets screwed to the transom **113**.

[0027] The boat **100** is equipped with an apparatus for towing a watersports participant. As shown in FIG. 1, the towing apparatus is integrated into a hardtop cover **142**, but other suitable towing apparatuses can be used, including, for example, the towers shown and described in U.S. Pat. Nos. 9,580,155 and 10,150,540, which are incorporated by reference herein in their entireties. The hardtop cover **142** is supported by two legs: a port leg **144** and a starboard leg **146**. The port leg **144** is attached to the boat **100** on the port side of the longitudinal centerline **102** of the boat **100** and the starboard leg **146** is attached to the boat **100** on the starboard side of the longitudinal centerline **102** of the boat **100**. Preferably, each of the port leg **144** and the starboard leg **146** is attached to an upper surface (such as gunwales) above the port-side hull **117** and the starboard-side hull **119**, respectively. The hardtop cover **142** is connected to an upper portion of each of the port leg **144** and the starboard leg **146** and spans the interior **120** of the boat **100** at a height suitable for passengers to pass underneath while standing. The hardtop cover **142** also extends forward and aft over a portion of the primary seating area **124** to provide shade. In addition, the hardtop cover **142** has a tow-line-attachment structure **148** at an aft portion of the hardtop cover **142**. This tow-line-attachment structure **148** can be used to connect a tow-line suitable for towing a watersports participant, such as a wake surfer. Any suitable tow-line-attachment structure can be used,

including, but not limited to, the integrated light and tow-line-attachment assembly disclosed in U.S. Pat. No. 6,539,886, which is incorporated by reference herein in its entirety.

[0028] The boat **100** can include other features for watersports. For example, although not depicted herein, the boat **100** can include the capability to add ballast. For example, the ballast systems shown and described in U.S. Patent Application Pub. No. 2024/0326957, which is incorporated by reference herein in its entirety, can be used in the boat **100**. The boat **100** can include other features and configurations for watersports, such as the operation of the propulsion system described in the '957 publication, or other surf devices, such as, for example, the port and starboard wake-modifying devices disclosed in U.S. Patent No. 8,833,286, which is incorporated by reference herein in its entirety. Other examples of suitable alternative surf devices are shown and described in U.S. Pat. Nos. 9,802,684 and 10,358,189 and the surf devices described in U.S. Patent Application Publication 2022/0089258, the disclosures of which are incorporated by reference herein in their entirety.

[0029] The boat **100** includes a propulsion system, including one or more propulsion units **150**. As depicted in FIG. **1**, the boat **100** can include a plurality of propulsion units **150**. The propulsion units **150** depicted in FIG. **1** are each outboards, but other types of propulsion systems can be used, including, for example, inboards, sterndrives, and jet drives. The propulsion unit **150** includes a motor **154** operatively coupled to a propulsor to drive the propulsor. In this embodiment, the motor **154** is a combustion engine, but other suitable motors can be used, including electric motors. The propulsor of this embodiment is a propeller **156**, but other suitable propulsors can be used, such as, for example, impellers in jet drives. The motor **154** is configured to drive (rotate) the propeller **156** through one or more drive shafts coupling the motor **154** to the propeller **156**. Rotating the propeller **156** drives the boat **100** through the water. To turn the boat **100**, an operator can turn a steering wheel at the control console **152** to rotate and angle the outboard motor (both the motor **154** and the propeller **156**) relative to the longitudinal centerline **102**.

[0030] As noted above, the boat **100** is equipped with an adjustable equipment storage rack **200**. The adjustable equipment storage rack **200** includes a cradle **212** (FIG. **2**) that is shaped and sized to hold a piece of equipment **10**, such as a piece of the watersport equipment. The piece of equipment **10** can be a board (for example, a paddleboard, a wakeboard, or a surfboard), a water ski, or a water mat. The piece of equipment **10** shown in the accompanying figures is a paddleboard. The adjustable equipment storage rack **200** can include one or more cradles **212**. A plurality of cradles **212**, such as two cradles **212**, can be used to support relatively large or long pieces of watersport equipment, like paddleboards. To maximize the space within the interior **120**, the cradles **212** of the adjustable equipment storage rack **200** are positioned on the exterior **130** of the hull **110**. Positioning the cradles **212** and the piece of equipment **10** stored therein with a majority of each cradle **212** and/or the piece of equipment **10** lower than the upper surface of the side of the deck or hull, such as lower than the top surface **134** of the perimeter **132**, has certain advantages to visibility and the like. Such a positioning provides better visibility for the driver to watch for other boaters or people in the water, as compared to other racks that have the piece of equipment stored above the top surface **134**. These other racks have a “truck topper” that makes it so the driver or others in the boat cannot see port aft or starboard aft past the port leg **144** or the starboard leg **146** in most situations. The piece of equipment effectively forms a wall, blocking visibility.

[0031] FIG. **2** shows a portion of the starboard-side hull **119** with the adjustable equipment storage rack **200** connected thereto. The adjustable equipment storage rack **200** can include one or more cradle assemblies. The adjustable equipment storage rack **200**, depicted in FIG. **2**, includes a plurality of cradle assemblies, including a first cradle assembly **201** and a second cradle assembly **202**. The plurality of cradle assemblies can be used to better support a large piece of equipment **10**, such as a paddle board or other boards. The following discussion will refer to the first cradle assembly **201**, but the discussion applies equally to other cradle assemblies of the plurality of

cradle assemblies, such as the second cradle assembly **202**.

[0032] The boat **100** includes a barrier **160** and the adjustable equipment storage rack **200** is mounted to the barrier **160**. The barrier **160** can be various barriers on the boat **100**, such as a bulkhead or other structure separating compartments within the boat **100**. The barrier **160** can be, for example, a barrier **160** that defines the interior **120** of the boat **100** and separates the interior **120** from the exterior **130** of the boat **100**. On a pontoon boat, for example, the barrier can be a fence. In the boat **100** depicted in FIG. 2, for example, the barrier **160** includes a portion of the hull such as one of the port-side hull **117** (FIG. 1) or the starboard-side hull **119**. In FIG. 1, the adjustable equipment storage rack **200** is shown as being mounted to the port-side hull **117**, but in FIG. 2, the adjustable equipment storage rack **200** is mounted to the starboard-side hull **119**. The barrier **160** includes an outward facing surface, which, as depicted in FIG. 2, is an exterior surface **162** of the barrier **160**. The barrier **160** also includes a surface opposite of the outward facing surface, such as an inward facing surface, which, as depicted in FIG. 2, is an interior surface **164**. The barrier **160** also includes a top surface **166**. The top surface **166** can be the top surface **134** of the perimeter **132** of the boat **100** as discussed above.

[0033] The first cradle assembly **201** includes a cradle **212** that is sized and shaped to support the piece of equipment **10**, such as by directly supporting the piece of equipment **10**. The cradle **212** can have various shapes, such as a J-shape (depicted), concave shapes, circular or arcuate shapes, U-shapes, V-shapes, or L-shapes. The cradle **212** can be sized and shaped depending upon the piece of watersport equipment to be stored therein. For example, the rack can be smaller for wakeboards or surfboards or larger for paddle boards. In another example, the rack can have a larger horizontal area when storing a rolled-up water mat or different shape. Likewise, although shown in the accompanying figures as storing only one piece of watersport equipment, the rack can have shapes that allow multiple pieces of watersport equipment to be stored.

[0034] The first cradle assembly **201** also includes a mounting bracket **214** connected to the cradle **212** to support the cradle **212**. The mounting bracket **214** connects the cradle **212** to the barrier **160**. The mounting bracket **214** can be attached to the barrier **160** at different locations. For example, the mounting bracket **214** can be attached to the top surface **166** of the barrier **160**. More specifically, the mounting bracket **214** can be directly attached to the top surface **166** of the barrier **160**. The first cradle assembly **201** can include a fastener **216**. The fastener **216** can engage with the mounting bracket **214** and secure the mounting bracket **214** to the boat **100**. The fastener **216** can secure the mounting bracket **214** to the barrier **160**, such as the top surface **166** of the barrier **160**.

[0035] FIGS. 3A and 3B are stern views of the barrier **160** and the adjustable equipment storage rack **200** attached thereto. The fastener **216** engages with the fastener **216** and secures the mounting bracket **214** to the barrier **160**, such as the top surface **166**, at a fastening position **222**. As noted above, the adjustable equipment storage rack **200** is adjustable. The adjustable equipment storage rack **200** can be positioned in a plurality of different positions. More specifically, the fastener **216** can engage with the mounting bracket **214** at a plurality of engagement positions **224**. At each engagement position **224** of the plurality of engagement positions **224**, the cradle **212** is a different distance from the fastening position **222**. For example, FIG. 3A shows the fastener **216** engaged with the mounting bracket **214** at a second engagement position **224b**, and FIG. 3B shows the fastener **216** engaged with the mounting bracket **214** at a fourth engagement position **224d**. The cradle **212** is farther from the fastening position **222** when the fastener **216** is engaged with the mounting bracket **214** at the second engagement position **224b** than when the fastener **216** is engaged with the mounting bracket **214** at the fourth engagement position **224d**.

[0036] The cradle **212** is preferably positioned to abut the barrier **160**, such as the exterior surface **162**. Such a positioning prevents vibration of the piece of equipment **10** in the cradle **212**. By having a plurality of engagement positions **224**, the cradle **212** can abut the exterior surface **162** evenly and accommodate barriers with different widths and different distances from the fastening position **222** to the exterior surface **162**.

[0037] FIGS. 4A and 4B are perspective views of the first cradle assembly **201**. FIG. 4A shows a front side of the first cradle assembly **201** and FIG. 4B shows a back side of the first cradle assembly **201**. As noted above, the first cradle assembly **201** includes a cradle **212**. The cradle **212** can have various shapes, such as the J-shape shown in FIGS. 4A and 4B. The cradle **212** includes a first leg and a second leg oriented transversely to the first leg. The first leg can be an upright leg **230** and the second leg can be a lateral leg **240**. The upright leg **230** includes an equipment-facing surface **232** and the lateral leg **240** also includes an equipment-facing surface **242**. The equipment-facing surfaces **232**, **242** can directly support the piece of equipment **10** (FIGS. 3A and 3B), such that the piece of equipment **10** abuts the equipment-facing surfaces **232**, **242** when placed in the cradle **212**. When these equipment-facing surfaces **232**, **242** directly support the piece of equipment **10**, padding, such as foam padding, can be placed on these surfaces to protect the piece of equipment **10** from damage.

[0038] The upright leg **230** also includes a back side surface **234**. The back side surface **234** is a surface opposite the equipment-facing surface **232** and faces away from the piece of equipment **10**. (FIGS. 3A and 3B) The back side surface **234** can face toward the barrier **160**, such as toward the exterior surface **162**. The upright leg **230** can be oriented in a generally vertical orientation when attached to the barrier **160**, and the lateral leg **240** can be oriented in a horizontal vertical orientation when attached to the barrier **160**. The upright leg **230** has an upper portion **236** and a lower portion **238**. The upper portion **236** can be an upper half, an upper third, or an upper quarter of the upright leg **230**. Similarly, the lower portion **238** can be a lower half, a lower third, or a lower quarter of the upright leg **230**. The lateral leg **240** is attached, such as directly attached, to the lower portion **238** of the upright leg **230**. More specifically, the lateral leg **240** is attached to a lower end of the upright leg **230**. The upright leg **230** and the lateral leg **240** can be a continuous piece, such as a continuous plate, and the lateral leg **240** can be formed by being bent relative to the upright leg **230**.

[0039] With the upright leg **230** and the lateral leg **240** positioned as discussed above and as shown in FIGS. 3A and 3B, the upright leg **230** can support a broad surface of the piece of equipment **10** and the lateral leg **240** can support a narrower surface of the piece of equipment **10**, when the piece of equipment **10** is located in the cradle **212**. As shown in FIG. 4A and 4B, the cradle **212** has a J-shape. The lateral leg **240** has a distal end **244**, which is an end of the lateral leg **240** distal from the upright leg **230**. The distal end **244** is bent or otherwise curved toward the piece of equipment **10** (FIGS. 3A and 3B). When mounted to the barrier **160**, the distal end **244** can be curved upward in the direction that the upright leg **230** extends from the lateral leg **240**. This upward bend of the distal end **244** helps retain and secure the bottom of the piece of equipment **10** in the cradle **212**.

[0040] The upright leg **230** includes an upper end **237**. The upper end **237** of the upright leg **230** can be curved or bent. The upper end **237** can be curved or bent in the direction that the back side surface **234** faces. As will be discussed further below, the mounting bracket **214** is attached to the back side surface **234** and extends from the back side surface **234**. The upper end **237** can be bent in the direction that the mounting bracket **214** extends. The bent portion of the upper end **237** can have, for example, an angle with the back side surface **234** that is from **110** degrees to **150** degrees. As shown in FIGS. 3A and 3B, for example, the upper end **237** can be bent toward the barrier **160**. When the cradle **212** is located on the exterior **130** of the boat **100**, the piece of equipment **10** can be placed into the cradle **212** by a person located in the interior **120** of the boat **100**. Bending the upper end **237** towards the interior **120** helps the person insert or remove the piece of equipment **10** from the cradle **212**.

[0041] To help protect the exterior surface **162** of the barrier **160**, such as the starboard-side hull **119**, one or more bumpers **218** can be attached, such as directly attached, to the back side surface **234** of the upright leg **230**. As shown in FIG. 4B, the bumpers **218** can be located on the lower portion **238** of the upright leg **230**. The bumpers **218** can be elastomeric bumpers, such as rubber bumpers attached by a suitable means to the lower portion **238**, such as by a fastener, like a screw,

or adhesive.

[0042] As illustrated in FIGS. 3A to 4B, the piece of equipment **10** can be further secured into the cradle **212** by one or more straps **250**. Each strap **250** can be engageable with the cradle to secure the piece of equipment **10** to the cradle **212**. The strap **250** can have one end attached to the distal end **244** of the lateral leg **240** and another end attached to the upper end **237** of the upright leg **230**. The strap **250** can comprise, at least in part, an elastomeric material, such as bungee cord **252**. In the depicted embodiment, the strap **250** also includes a sheath **254**, such as a canvas sheath. The bungee cord **252** attaches to or extends through the sheath **254**. With the ends of the strap **250** attached as discussed above, the strap **250** compresses the piece of equipment **10** against the cradle **212** and secures the piece of equipment **10** in the cradle **212**.

[0043] The strap **250** can be detachably connected to the cradle **212**, such as detachably connected to the upper end **237** of the upright leg **230**. The upper end **237** of the upright leg **230** can include a slot **256**, and a portion of the strap **250**, such as a portion of the bungee cord **252**, extends through the slot **256**. A cord stop **258** can be connected to the bungee cord **252**. The cord stop **258** can be a cord lock, such as a cord lock with a spring-loaded clasp that grips the cord. The cord stop **258** can be sized to have at least one dimension larger than the slot **256** to prevent the cord stop **258** from moving through the slot **256**, thus securing the bungee cord **252** in the slot **256**. The elastic force of the strap **250** pulls the cord stop **258** against the back side surface **234** to hold the cord stop **258** in place. When a cord lock is used, the length of the strap can be adjusted to secure pieces of equipment **10** (e.g., boards) of different sizes and apply the elastic force to the piece of equipment **10** and the cord stop **258**. To release the strap **250**, the user can pull on the end of the strap **250** with the cord stop **258** to move the cord stop **258** away from the back side surface **234** and then slide the bungee cord **252** out of the slot **256**.

[0044] As shown in FIGS. 4A and 4B, the mounting bracket **214** is connected to the cradle **212**, such as attached to the cradle **212**. More specifically, the mounting bracket **214** is attached to the back side surface **234**, such as directly attached to the back side surface **234** of the upright leg **230**. The mounting bracket **214** is attached to the upper portion **236** of the upright leg **230** to position the cradle **212** with a majority of the cradle **212** lower than the top surface **166** of the exterior surface **162** (see FIGS. 3A and 3B). The mounting bracket **214** can be formed as a plate (i.e., a mounting bracket plate **262**) with a mounting bracket flange **264** formed on one end of the mounting bracket plate **262**. The mounting bracket flange **264** can be directly attached to the back side surface **234** of the upright leg **230** using, for example, fasteners. The mounting bracket flange **264** can be attached to the back side surface **234** at a plurality of different heights to accommodate different heights of the barrier **160**. For example, the cradle **212** is preferably positioned high enough that the lateral leg **240** is out of the water when positioned on the exterior **130** of the boat **100**.

[0045] FIG. 5 is a detail view of the mounting bracket **214**. To help protect the barrier **160**, such as the top surface **166** of the barrier **160**, a pad **266** can be located on an underside of the mounting bracket plate **262**. The pad **266** can be, for example, a plastic pad, such as high-density polyethylene.

[0046] The mounting bracket **214** shown in FIG. 5 includes a slot **272**. The slot **272** is an elongated slot that is elongated in a longitudinal direction L of the slot **272**. The slot **272** also includes a width direction W, which is a direction that is perpendicular to the longitudinal direction L of the slot **272**. The mounting bracket **214** is also oriented in the longitudinal direction L and the width direction W. The fastener **216** can engage with the mounting bracket **214** by being located in the slot **272**. For example, the slot **272** extends through the mounting bracket plate **262** and the pad **266** and the fastener **216** can extend through the slot **272** to engage with a receiver **310** (FIG. 6) located in the barrier **160** (FIG. 6), thus connecting the mounting bracket **214** to the barrier **160**. As will be discussed further below, the fastener **216** can slide within the slot when the mounting bracket **214** is moved between the plurality of engagement positions **224**.

[0047] The mounting bracket **214** includes a plurality of openings **270**. Each opening of the

plurality of openings **270** can be a hole, such as a blind hole or a through hole. Some of the plurality of openings **270** are in a first set of openings **274**, and some of the plurality of openings **270** are in a second set of openings **276**. The openings of the first set of openings **274** are arrayed in a linear array on one side of the slot **272**, and the openings of the second set of openings **276** are arrayed in a linear array on an opposite side of the slot **272** in the width direction **W**. The openings of the first set of openings **274** and the openings of the second set of openings **276** are arranged to form corresponding pairs of openings **278**. Each pair of openings **278** includes a first opening **274x** from the first set of openings **274** and a second opening **276x** from the second set of openings **276**. The first opening **274x** is thus located on one side of the slot **272** and the second opening **276x** is located on an opposite side of the slot **272** in the width direction **W**. Each pair of openings **278** corresponds to one engagement position of the plurality of engagement positions **224**. Similarly, each opening of the first set of openings **274** corresponds to one engagement position of the plurality of engagement positions **224**, and each opening of the second set of openings **276** corresponds to another engagement position of the plurality of engagement positions **224**. For example, as depicted in FIG. 5, the mounting bracket **214** includes six engagement positions and corresponding pairs of openings **278**. These different positions, positions one to six, are shown in the figures with the reference character a to f appended to reference numerals **224** and **274** to indicate the first through sixth position, respectively.

[0048] FIG. 6 is a detail view of the barrier **160** with the first cradle assembly **201** and the second cradle assembly **202** is removed. The barrier **160** includes one or more receivers **310** located in the barrier **160**. In FIG. 6, the barrier **160** includes two receivers, one for the first cradle assembly **201** (FIG. 2) and one for the second cradle assembly **202** (FIG. 2). The receiver **310** can be embedded in the barrier **160**, such as in the top surface **166** of the barrier **160**. The receiver **310** includes a receiver bore **312** to receive the fastener **216**. The receiver **310** can also include a receiver flange **314** to assist in attaching the receiver **310** to the top surface **166** of the barrier **160**.

[0049] FIGS. 7A and 7B show a fastener assembly **300** that can be used with the adjustable equipment storage rack **200**. The fastener assembly **300** includes the receiver **310** and a quick-release fastener **320** that can be used as the fastener **216** discussed herein. Although described as a quick-release fastener **320**, the fastener **216** can be another type of fastener such as other bolts, pins, and the like. FIG. 7A shows the quick-release fastener **320** connected to the receiver **310** and FIG. 7B shows the quick-release fastener **320** removed from the receiver **310**.

[0050] The receiver **310** shown in FIGS. 7A and 7B includes a receiver shaft **316** extending from the receiver flange **314**. The receiver bore **312** (FIG. 6) is formed in the receiver shaft **316**. The receiver shaft **316** can include threads formed on an exterior thereof, and a receiver nut **318** can be threaded onto the receiver shaft **316** to secure the portion of the barrier **160** with the top surface **166** between the receiver nut **318** and the receiver flange **314**.

[0051] The quick-release fastener **320** includes a fastener shaft **322**. The fastener shaft **322** engages with the receiver **310**. More specifically, the fastener shaft **322** is inserted into the receiver bore **312**. The quick-release fastener **320** includes a catch **324** movable between a latched position and an unlatched position. The latched position is depicted in FIG. 7B with the catch **324** extending from an outer surface of the fastener shaft **322**. In the unlatched position, the catch **324** can be retracted into the fastener shaft **322**. The catch **324** of the quick-release fastener **320** can be a detent, for example. When in the latched position, the catch **324** engages with, for example, a recess or a slot (not shown) formed on an inner surface of the receiver bore **312** to prevent the fastener shaft **322** from being withdrawn from the receiver bore **312**. To withdraw the fastener shaft **322** from the receiver bore **312**, the catch **324** is moved to the unlatched position. A biasing member, such as a spring, can be used to exert a biasing force on the catch **324** to push the catch **324** toward the latched position and hold the catch **324** in the latched position.

[0052] The quick-release fastener **320** can include a release **326**. The release **326** is operably connected to the catch **324** to allow the catch **324** to move from the latched position to the

unlatched position when the release **326** is activated. Here, the release **326** is a button located on an upper surface of the quick-release fastener **320**. The release **326** can be activated by a user pressing the button, which is connected to the catch **324** to allow the catch to retract.

[0053] The quick-release fasteners **320** provide a secure connection, but allow for easy, tool-free engagement or disengagement to remove the adjustable equipment storage rack **200** when not in use. While the quick-release fastener **320** has been discussed herein as a button-activated spring detent quick-release fastener, other quick-release fasteners can be used, such as, for example, a lever-based quick-release fastener, a cam lock quick-release fastener, or a ratchet quick-release fastener. A lever-based quick-release fastener can use a lever that operates a cam or spring-loaded mechanism to lock and release components. The lever, when activated, either tightens or loosens the fastener, allowing the components to be securely attached or easily detached without the need for tools. The cam lock quick-release fastener utilizes a rotating cam mechanism to lock or unlock components. The cam can include a slanted surface that, when rotated, such as by a knob or handle, either pulls the components together or pushes them apart. The ratchet quick-release fastener utilizes a ratcheting system to incrementally lock parts into place. The ratchet mechanism features a set of teeth that engage with corresponding teeth or gears to secure the components. The fastener is locked by rotating or pulling the mechanism until the teeth are engaged and the release occurs when the ratchet system is disengaged, usually by pulling a lever or pressing a button to release the teeth.

[0054] The quick-release fastener **320** also includes a shank **330**. The shank **330** is the portion of the quick-release fastener **320** that is positioned in and slides within the slot **272** (FIG. 5) of the mounting bracket **214** (FIG. 5). The shank **330** is located between an upper flange **332** and a lower flange **334**. The upper flange **332** and the lower flange **334** are located on either side of the mounting bracket plate **262** and pad **266**, if used, as shown in FIGS. 8A to 8D. The upper flange **332** and the lower flange **334** prevent the quick-release fastener **320** from being removed from the slot **272** when the position of the first cradle assembly **201** is being adjusted, as will be discussed further below.

[0055] The upper flange **332** includes one or more projections. The upper flange **332** shown in FIG. 7B includes a first projection **336** and a second projection **338**. The first projection **336** and the second projection **338** extend downward from the upper flange **332**. The first projection **336** is engageable with the first opening **274x** (FIG. 5) of each pair of openings **278** (FIG. 5) and the second projection **338** is engageable with the second opening **276x** (FIG. 5) of each pair of openings **278**.

[0056] FIGS. 8A to 8D illustrate the process of adjusting the position of the first cradle assembly **201**. The cradle **212** is positionable in a plurality of cradle positions. At each different position of the plurality of cradle positions, the cradle **212** is a different distance from the barrier **160**. Each engagement position **224** of the plurality of engagement positions **224** corresponds to one cradle position of the plurality of cradle positions, and engaging the quick-release fastener **320** with the mounting bracket **214** at one of the plurality of engagement positions **224** locates the cradle **212** at a corresponding one of the plurality of cradle positions. To move the cradle **212** from one cradle position to another, the quick-release fastener **320** is moved from one engagement position **224** to another. As will be illustrated in FIGS. 8A to 8D, for example, the quick-release fastener **320** will be moved from the second engagement position **224b** to the fourth engagement position **224d**.

[0057] FIG. 8A shows the mounting bracket **214** with the quick-release fastener **320** engaged with the mounting bracket **214** in the second engagement position **224b**, such as when the adjustable equipment storage rack **200** is positioned as depicted in FIG. 3A. Here, the fastener shaft **322** is located in the receiver bore **312** (FIG. 6) with the catch **324** in the latched position to secure the quick-release fastener **320** and the mounting bracket **214** to the barrier **160**. The first projection **336** is engaged with the first opening **274x** (FIG. 8B) of the second pair of openings **278b** and the second projection **338** is engaged with the second opening **276x** of the second pair of openings

278b.

[0058] As shown in FIG. 8B, the release **326** is activated to allow the catch **324** to move from the latched position to the unlatched position and the fastener shaft **322** to be withdrawn, at least in part, from the receiver bore **312**. The quick-release fastener **320** can be raised until the lower flange **334** (FIG. 7B) contacts the underside surface of the mounting bracket **214**. The distance between the upper flange **332** and the lower flange **334** can be set to allow the first projection **336** and the second projection **338** to disengage from the first opening **274x** and the second opening **276x**, respectively, of the second pair of openings **278b**.

[0059] As shown in FIG. 8C, with the quick-release fastener **320** in the position discussed above, the mounting bracket **214** can be moved relative to the quick-release fastener **320**, such as by sliding. The quick-release fastener **320** slides within the slot **272** as the mounting bracket **214** moves, and the fastener shaft **322** can remain partly within the receiver bore **312**. When the cradle **212** is in the desired cradle position, the quick-release fastener **320** can be reinserted fully into the receiver bore **312** and the first projection **336** and the second projection **338** engaged with the first opening **274x** and the second opening **276x**, respectively, of the fourth pair of openings **278d**, as shown in FIG. 8D. The release **326** is then deactivated, securing the quick-release fastener **320** within the receiver **310** by the catch **324** moving to the latched position. With the quick-release fastener **320** engaged with the mounting bracket **214** in the fourth pair of openings **278d**, the quick-release fastener **320** is located at the fourth engagement position **224d** to position the adjustable equipment storage rack **200** as depicted in FIG. 3B.

[0060] FIGS. 9A and 9B are perspective views of another cradle assembly, referred to herein as a third cradle assembly **203**, that can be used as the cradle assemblies of the adjustable equipment storage rack **200**. The third cradle assembly **203** is similar to the first cradle assembly **201** discussed above and that discussion applies here. The same reference numerals are used for features of the third cradle assembly **203** that are the same or similar to the corresponding features discussed above. FIG. 9A shows a front side of the third cradle assembly **203** and FIG. 9B shows a back side of the third cradle assembly **203**. The cradle **212** of the third cradle assembly **203** has a different shape than the cradle **212** of the first cradle assembly **201**. In the first cradle assembly **201**, discussed above, the upper end **237** of the upright leg **230** is bent, but in the third cradle assembly **203**, this feature is omitted and the equipment-facing surface **232** of the upright leg **230** continues to extend upward without a bend or curve in the direction of the mounting bracket **214**. [0061] Each cradle assembly in the adjustable equipment storage rack **200** can be the same or different. For example, the cradles **212** of each of the cradle assemblies used in the adjustable equipment storage rack **200** can be shaped differently. For example, a cradle assembly having one shape can be the first cradle assembly **201** and another having a different shape can be the third cradle assembly **203**. Other shapes and positions can be used to accommodate different pieces of equipment.

[0062] FIGS. 10A and 10B are perspective views of another cradle assembly, referred to herein as a fourth cradle assembly **204**, that can be used as the cradle assemblies of the adjustable equipment storage rack **200**. The fourth cradle assembly **204** is similar to the first cradle assembly **201** discussed above and that discussion applies here. The same reference numerals are used for features of the fourth cradle assembly **204** that are the same or similar to the corresponding features discussed above. FIG. 10A shows a front side of the fourth cradle assembly **204**, and FIG. 10B shows a back side of the fourth cradle assembly **204**.

[0063] The fourth cradle assembly **204** includes a cradle **280** that is shaped to support a different piece of equipment, such as a buoyant water mat. Such water mats can be stored in a rolled or cylindrical configuration. Here, the cradle **280** has an arcuate shape with an equipment-facing surface **282** that is also arcuate. The cradle **280** can have an arcuate shape that extends greater than **90** degrees. The cradle **280** is attached to the upright leg **230**.

[0064] The piece of equipment **10** can be secured in the cradle **280** using a strap **284**. As noted

above, different straps can be used, and the strap **284** depicted in FIGS. **10A** and **10B** is a nylon strap. The water mat can be placed against the cradle **280** and then the strap **284** can be wrapped around the water mat and secured by closing a clasp **286** of the strap **284** to connect each end of the straps **284**. The strap **284** can be positioned behind the cradle **280**, such as between the cradle **280** and the upright leg **230**, to retain the strap **284** within the fourth cradle assembly **204**.

[0065] FIG. **11** is a detail view of a clasp **286** of the strap **284**. Different clasps can be used, but the clasp **286** depicted in FIG. **11** is a side-release buckle that can have two parts: one part with a prong (or tongue) and the other with a slot, and the two parts snap together. The side-release buckle can be quickly released by pressing the sides of the buckle. The nylon strap is threaded through each part of the side-release buckle, creating a loop that allows for easy length adjustment. By pulling the free end of the strap through the buckle, the length of the strap **284** can be increased or decreased to tighten the strap **284** around the water mat and adjust for mats of different sizes.

[0066] As used herein, directional terms forward (fore), aft, inboard, and outboard have their commonly understood meaning in the art. Relative to the boat, forward is a direction toward the bow and aft is a direction toward the stern. Likewise, inboard is a direction toward the center of the boat and outboard is a direction away from it.

[0067] The terms “coupled,” “fixed,” “attached,” “connected,” and the like refer to both direct coupling, fixing, attaching, or connecting, as well as indirect coupling, fixing, attaching, or connecting, through one or more intermediate components or features, unless otherwise specified herein.

[0068] Although this invention has been described with respect to certain specific exemplary embodiments, many additional modifications and variations will be apparent to those skilled in the art in light of this disclosure. It is, therefore, to be understood that this invention may be practiced otherwise than as specifically described. Thus, the exemplary embodiments of the invention should be considered in all respects to be illustrative and not restrictive and the scope of the invention to be determined by any claims supportable by this application and the equivalents thereof, rather than by the foregoing description.

Claims

1. An adjustable equipment storage rack for a boat, the adjustable equipment storage rack comprising: a cradle for supporting a piece of equipment; and a mounting bracket connected to the cradle to support the cradle, the mounting bracket having a plurality of engagement positions for a fastener to engage with the mounting bracket and to secure the mounting bracket to the boat at a fastening position, wherein, at each engagement position of the plurality of engagement positions, the cradle is a different distance from the fastening position.
2. The adjustable equipment storage rack of claim 1, further comprising a strap engageable with the cradle to secure the piece of equipment to the cradle.
3. The adjustable equipment storage rack of claim 1, wherein the cradle includes a first leg and a second leg oriented transversely to the first leg and connected to the first leg at a lower portion thereof.
4. The adjustable equipment storage rack of claim 3, wherein the mounting bracket is attached to a back side surface of the first leg and projects therefrom, and an upper end of the first leg is bent in the direction that the mounting bracket projects.
5. The adjustable equipment storage rack of claim 1, wherein the mounting bracket includes a plurality of openings, and each opening corresponds to one engagement position of the plurality of engagement positions and is engageable with a projection to position the bracket at a corresponding engagement position of the plurality of engagement positions.
6. The adjustable equipment storage rack of claim 5, further comprising the fastener, the fastener being a quick-release fastener.

7. The adjustable equipment storage rack of claim 5, further comprising the fastener, wherein the mounting bracket includes a slot, and the fastener is located in the slot and is capable of sliding within the slot when the mounting bracket is moved between the plurality of engagement positions.
8. The adjustable equipment storage rack of claim 7, wherein the mounting bracket includes a plurality of openings, and each opening corresponds to one engagement position of the plurality of engagement positions, and wherein the fastener includes a projection engageable with an opening of the plurality of openings to position the bracket at a corresponding engagement position of the plurality of engagement positions.
9. The adjustable equipment storage rack of claim 7, wherein the slot includes a longitudinal direction and a width direction perpendicular to the longitudinal direction and the slot is elongated in the longitudinal direction, wherein the mounting bracket includes a plurality of pairs of openings, each pair of openings corresponds to one engagement position of the plurality of engagement positions, each pair of openings includes a first opening and a second opening, and the first opening is located on one side of the slot and the second opening is located on an opposite side of the slot in the width direction, and wherein the fastener includes a first projection engageable with the first opening and a second projection engageable with the second opening to position the bracket at a corresponding engagement position of the plurality of engagement positions.
10. A boat comprising: a barrier; and an adjustable equipment storage rack mounted to the barrier, the adjustable equipment storage rack including a cradle assembly, the cradle assembly including: a cradle for supporting a piece of equipment; and a mounting bracket connecting the cradle to the barrier, the mounting bracket being positionable in a plurality of cradle positions, wherein, at each position of the plurality of cradle positions, the cradle is a different distance from the barrier.
11. The boat of claim 10, wherein the adjustable equipment storage rack further includes a fastener engageable with the mounting bracket at a plurality of engagement positions, each engagement position of the plurality of engagement position corresponds to one cradle position of the plurality of cradle positions, and engaging the fastener with the mounting bracket at one of the plurality of engagement positions locates the cradle at a corresponding one of the plurality of cradle positions.
12. The boat of claim 11, wherein the barrier includes a receiver and the fastener is engageable with the receiver to connect the mounting bracket to the barrier.
13. The boat of claim 12, wherein the receiver includes a bore, and the fastener includes a fastener shaft that engages with the receiver when located in the bore, and one of the receiver and the fastener shaft includes a detent to lock the fastener shaft in the bore.
14. The boat of claim 10, wherein the barrier has a top surface and the mounting bracket is attached to the top surface of the barrier.
15. The boat of claim 14, wherein the majority of the cradle is located lower than the top surface of the barrier.
16. The boat of claim 10, wherein the barrier defines an interior of the boat and separates the interior from an exterior of the boat.
17. The boat of claim 16, wherein the cradle is located on the exterior of the boat.
18. The boat of claim 16, further comprising a hull, the hull including a port-side hull and a starboard-side hull, the barrier being one of the port-side hull or the starboard-side hull.
19. The boat of claim 16, wherein the cradle assembly is a first cradle assembly and the adjustable equipment storage rack includes a plurality of cradle assemblies, including the first cradle assembly and a second cradle assembly, the second cradle assembly including: a cradle for supporting the piece of equipment; and a mounting bracket connecting the cradle to the barrier, the mounting bracket being positionable in a plurality of cradle positions, wherein, at each position of the plurality of cradle positions, the cradle is a different distance from the barrier.
20. The boat of claim 19, wherein the cradle of each of the first cradle assembly and the second cradle assembly has the same shape.
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